EXP NO:1 Image handling and pixel transformation using Open CV

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Step1: Load an image from your local directory and display it.

In [13…
```
import cv2
import matplotlib.pyplot as plt
```

In [13…
```
# Read the image using OpenCV
img = cv2.imread('vaishu.jpeg', cv2.IMREAD_COLOR)
```

In [13…
```
# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color orde
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [13…
```
# Display the image using Matplotlib
plt.imshow(img_rgb, cmap='viridis')  # You can change 'viridis' to anothe
plt.title("Original Image")
plt.axis('off')  # Removes axis ticks and labels
plt.show()
```



Step2: o Draw a line from the top-left to the bottom-right of the image. o Draw a circle at the center of the image. o Draw a rectangle around a specific region of interest in the image. o Add the text "OpenCV Drawing" at the top-left corner of the image.

Draw a line from the top-left to the bottom-right of the image

In [84…
```
# Load the image
image = cv2.imread('vaishu.jpeg')
```

In [85…
```
# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color orde
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [86…
```
img_rgb.shape
```

Out[86]:
```
(1600, 900, 3)
```

In [87…
```
# Draw a line from top-left to bottom-right
line_img = cv2.line(img_rgb, (0, 0), (768, 600), (255, 0, 0), 2) # cv2.li
```

In [88…
```
plt.imshow(line_img, cmap='viridis')
plt.title("Image with Line")
plt.axis('off')
plt.show()
```



Draw a circle at the center of the image.

In [89…
```
# Load the image
image = cv2.imread('vaishu.jpeg')

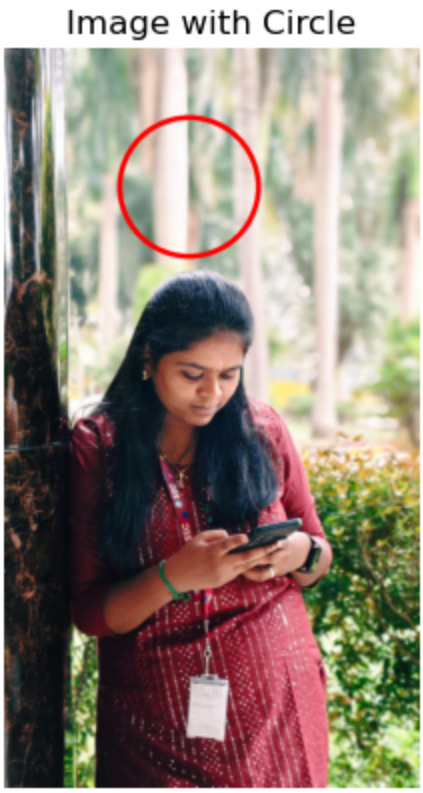
# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color orde
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [90…
```
img_rgb.shape
```

Out[90]:
```
(1600, 900, 3)
```

In [91…
```
circle_img = cv2.circle(img_rgb,(400,300),150,(255,0,0),10) # cv2.circle(
```

In [92…
```
plt.imshow(circle_img, cmap='viridis')
plt.title("Image with Circle")
plt.axis('off')
plt.show()
```



Draw a rectangle around the whole image

In [93…
```
# Load the image
image = cv2.imread('vaishu.jpeg')

# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color orde
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [94…
```
img.shape
```

Out[94]:
```
(1600, 900, 3)
```

In [95…
```
# Draw a rectangle around the Whole image
rectangle_img = cv2.rectangle(img_rgb, (0, 0), (768, 600), (0, 0, 255), 1
```

In [96…
```
plt.imshow(rectangle_img, cmap='viridis')
plt.title("Image with Rectangle")
plt.axis('off')
plt.show()
```



Add the text "OpenCV Drawing" at the top-left corner of the image.

In [16…
```
# Load the image
image = cv2.imread('vaishu.jpeg')

# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color orde
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

In [16…
```
# Add text to the image
text_img = cv2.putText(img_rgb, "vaishnavi s", (10, 30), cv2.FONT_HERSHEY
```

In [16…
```
plt.imshow(text_img, cmap='viridis')
plt.title("vaishnavi s")
plt.axis('off')
plt.show()
```



Step3: o Convert the image from RGB to HSV and display it. o Convert the image from RGB to GRAY and display it. o Convert the image from RGB to YCrCb and display it. o Convert the HSV image back to RGB and display it.

In [10…
```
# Load the image
image = cv2.imread('vaishu.jpeg')
```

In [10…
```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

In [10…
```
# Original RGB Image
plt.imshow(image_rgb)
plt.title("Original RGB Image")
plt.axis("off")
```

Out[102]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```

Original RGB Image



In [10…
```
# Convert RGB to HSV
image_hsv = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2HSV)
```

In [10…
```
# HSV Image
plt.imshow(image_hsv)
plt.title("HSV Image")
plt.axis("off")
```

Out[104]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```



In [10…
```
# Convert RGB to GRAY
image_gray = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2GRAY)
```

In [10…
```
# Grayscale Image
plt.imshow(image_gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")
```

Out[106]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```



In [10…
```
# Convert RGB to YCrCb
image_ycrcb = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2YCrCb)
```

In [10…
```
# YCrCb Image
plt.imshow(image_ycrcb)
plt.title("YCrCb Image")
plt.axis("off")
```

Out[108]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```

YCrCb Image



In [10…
```
# Convert HSV back to RGB
image_hsv_to_rgb = cv2.cvtColor(image_hsv, cv2.COLOR_HSV2RGB)
```

In [11…
```
plt.imshow(image_hsv_to_rgb)
plt.title("HSV to RGB Image")
plt.axis("off")
```

Out[110]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```



Step4: o Access and print the value of the pixel at coordinates (100, 100). o Modify the color of the pixel at (200, 200) to white.

In [11…
```
# Modify a block of pixels (300x300) to white, starting from (200, 200)
image[200:500, 200:500] = [255, 255, 255]  # Rows: 200-499, Columns: 200-
```

In [11…
```
# Convert BGR to RGB for displaying with Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

In [11…
```
# Display the modified image
plt.imshow(image_rgb)
plt.title("Image with 300x300 White Block")
plt.axis("off")
plt.show()
```



Step5: o Resize the original image to half its size and display it.

In [11…
```
# Load the image
image = cv2.imread('vaishu.jpeg')
```

In [11…
```
image.shape
```

Out[115]:
```
(1600, 900, 3)
```

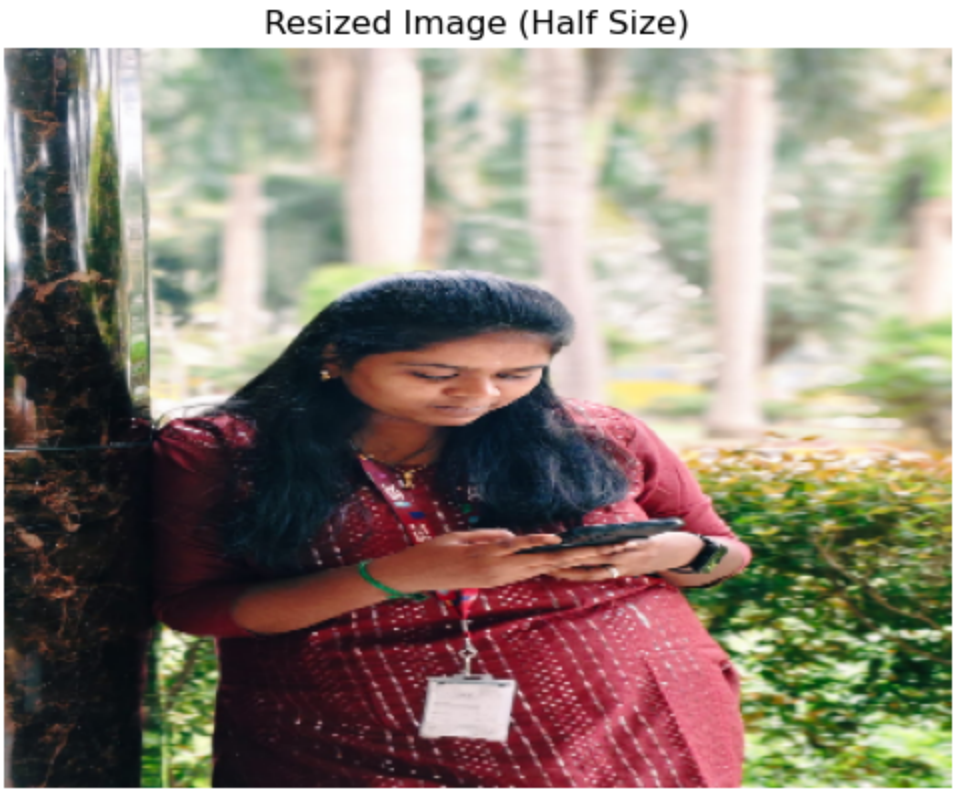
In [11…
```
# Resize the image to half its size
resized_image = cv2.resize(image, (768 // 2, 600 // 2))  # (new_width, ne
```

In [11…
```
# Convert BGR to RGB for displaying with Matplotlib
resized_image_rgb = cv2.cvtColor(resized_image, cv2.COLOR_BGR2RGB)
```

In [11…
```
resized_image_rgb.shape
```

Out[118]:
```
(300, 384, 3)
```

In [11…
```
# Display the resized image
plt.imshow(resized_image_rgb)
plt.title("Resized Image (Half Size)")
plt.axis("off")
plt.show()
```



Step6: o Crop a region of interest (ROI) from the image (e.g., a 100x100 pixel area starting at (50, 50)) and display it.

In [12…
```
# Load the image
image = cv2.imread('vaishu.jpeg')
```

In [12…
```
image.shape
```

Out[121]:
```
(1600, 900, 3)
```

In [12…
```
# Crop a 300x300 region starting from (50, 50)
roi = image[50:350, 50:350]  # Rows: 50-349, Columns: 50-349
```

In [12…
```
# Convert BGR to RGB for displaying with Matplotlib
roi_rgb = cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)
```

In [12…
```
# Display the cropped region (ROI)
plt.imshow(roi_rgb)
plt.title("Cropped Region of Interest (ROI)")
plt.axis("off")
plt.show()
```



Step7: o Flip the original image horizontally and display it. o Flip the original image vertically and display it.

In [12…
```
# Load the image
image = cv2.imread('vaishu.jpeg')
```

In [12…
```
# Flip the image horizontally (left-right)
flipped_horizontally = cv2.flip(image, 1)
```

In [12…
```
# Convert BGR to RGB for displaying with Matplotlib
flipped_horizontally_rgb = cv2.cvtColor(flipped_horizontally, cv2.COLOR_B
```

In [12…
```
# Horizontal flip
plt.imshow(flipped_horizontally_rgb)
plt.title("Flipped Horizontally")
plt.axis("off")
```

Out[128]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```



In [12…
```
# Flip the image vertically (up-down)
flipped_vertically = cv2.flip(image, 0)
```

In [13…
```
# Convert BGR to RGB for displaying with Matplotlib
flipped_vertically_rgb = cv2.cvtColor(flipped_vertically, cv2.COLOR_BGR2R
```

In [13…
```
# Vertical flip
plt.imshow(flipped_vertically_rgb)
plt.title("Flipped Vertically")
plt.axis("off")
```

Out[131]:
```
(np.float64(-0.5), np.float64(899.5), np.float64(1599.5), np.float6
4(-0.5))
```



Step8: o Save the final modified image to your local directory.

In [ ]: